

Akshat Kumar Sinha

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SUMMARY

Computer Science undergraduate specializing in Data Science with software engineering internship experience building enterprise applications using React, TypeScript, Python, and REST APIs. Hands-on experience in machine learning, natural language processing, computer vision, financial data analysis, and fraud detection. NPTEL-certified in Natural Language Processing, with experience training classification models, performing feature engineering, evaluating model performance, and working with SQL, PostgreSQL, MongoDB, OpenCV, Pandas, and Scikit-learn.

EXPERIENCE

Software Development Engineer Intern

Jun 2026 – Present

Routa Digital India

- Accelerated delivery of **6+ enterprise security modules** by developing reusable React and TypeScript components integrated with REST APIs, reducing average feature development time by approximately **30%**.
- Reduced server load by approximately **40%** on high-frequency reporting endpoints by implementing batch processing for bulk user operations and replacing per-record API calls with paginated and chunked request pipelines.
- Improved page load performance by approximately **35%** across three data-intensive dashboard modules by identifying excessive API requests and coordinating with backend engineers to consolidate REST operations.
- Implemented role-based access control across **6+ security modules** using permission-scoped components, dynamic user-group management, and role-conditional workflows for multi-level enterprise users.
- Collaborated with frontend and backend developers to debug API integration issues, optimize data flows, and deliver maintainable software features for a production cyber-risk management platform.

PROJECTS

Fraud Transaction Detection using Machine Learning

Python, Scikit-learn, Pandas, NumPy

- Trained supervised machine learning models to identify potentially fraudulent financial transactions using transaction-level data and engineered numerical features.
- Performed data preprocessing, missing-value handling, class-distribution analysis, feature selection, model training, and hyperparameter tuning.
- Evaluated classification models using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC to account for imbalanced fraud datasets.
- Authored an unpublished research paper documenting the problem statement, literature review, methodology, experimental setup, model evaluation, results, and limitations.

Banking Market Intelligence Automation

Python, Pandas, Financial APIs

- Reduced manual financial data collection time by approximately **80%** by building an automated Python pipeline that retrieves, cleans, normalizes, and analyzes market data for **10+ banking stocks**.
- Generated structured Excel reports containing stock metrics, historical trends, comparative analysis, and summarized market insights.
- Applied Pandas-based data transformation, exception handling, time-series processing, and automated report generation to create a reusable analytical workflow.

FI Calculations

JavaScript, HTML, CSS, CSV Processing

- Developed and deployed a financial calculator supporting **5+ investment metrics**, including SIP growth, compound interest, investment returns, loan amortization, and financial independence calculations.
- Designed modular calculation logic using object-oriented programming principles and implemented CSV-based data processing for structured financial input.

Student Access Portal

JavaScript, Node.js, JWT, RBAC

- Built a three-role campus access platform for students, mentors, and administrators with JWT-based authentication and route-level authorization.

- Implemented role-specific dashboards and approval workflows to centralize temporary access requests, scheduling, and administrative resource management.

Self-Hosted Home Server

Ubuntu Server, CasaOS, Tailscale, Linux

- Configured and maintained a self-hosted Ubuntu Server with CasaOS and Tailscale, enabling secure remote access to locally hosted applications and services.
- Gained hands-on experience with Linux administration, networking, Docker-based services, system monitoring, service configuration, and troubleshooting.

RESEARCH EXPERIENCE

Machine Learning-Based Fraud Transaction Detection

2026

- Investigated machine learning techniques for detecting fraudulent financial transactions using classification algorithms and transaction datasets.
- Conducted exploratory data analysis, feature engineering, model comparison, and evaluation with a focus on imbalanced classification.
- Wrote a research paper covering related work, proposed methodology, implementation workflow, experimental results, and possible future improvements.
- Paper is currently unpublished and was completed as an academic research project.

EDUCATION

2023 – 2027	B.Tech, Computer Science and Engineering <i>Data Science Specialization</i> SRM Institute of Science and Technology	CGPA: 8.19 / 10
2023	Class 12th, CBSE	64.6%
2021	Class 10th, CBSE	71.8%

CERTIFICATIONS

Natural Language Processing – NPTEL

- Completed an NPTEL-certified course covering text preprocessing, tokenization, stemming, lemmatization, language modelling, text classification, topic modelling, word representations, and NLP evaluation techniques.

SKILLS

Languages	Python, C, C++, Java, JavaScript, TypeScript, Kotlin, SQL
AI and ML	Machine Learning, Supervised Learning, Classification, Regression, Feature Engineering, Data Preprocessing, Model Evaluation, Hyperparameter Tuning
NLP	Tokenization, Stemming, Lemmatization, Bag of Words, TF-IDF, Text Classification, Sentiment Analysis, Topic Modelling, Word Embeddings, Transformer Fundamentals
Computer Vision	OpenCV, Image Preprocessing, Thresholding, Edge Detection, Contour Detection, Image Classification Fundamentals
Libraries	Pandas, NumPy, Scikit-learn, OpenCV, Matplotlib
Databases	SQL, PostgreSQL, MongoDB, Relational Database Design, Queries, Joins, Indexing, Aggregation
Frontend	React.js, TypeScript, JavaScript, HTML, CSS, Tailwind CSS
Backend	Node.js, REST APIs, JWT Authentication, Role-Based Access Control, Batch Processing
Core CS	Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
Tools	Git, GitHub, VS Code, Linux, Ubuntu Server, Docker, CasaOS, Tailscale, Netlify
DSA Practice	Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Sorting, Searching, Recursion, and regular LeetCode problem solving